

Fondamenti di Analisi dei Dati

from data analysis to predictive techniques

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Causal Analysis

What has to be true before the word "causes" can be used at all

Correlation Is Not Causation — So What?

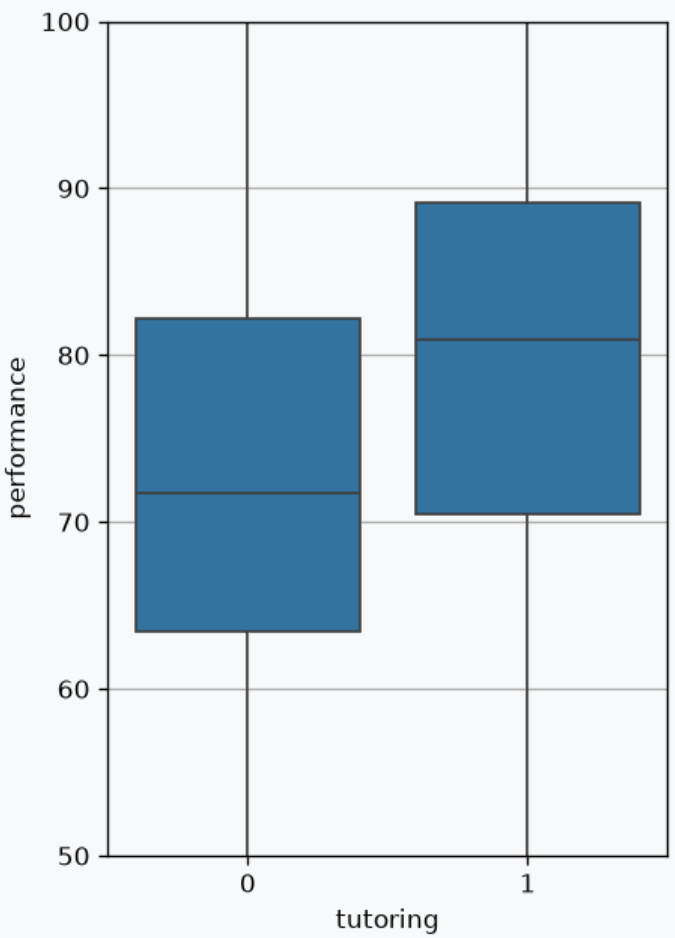
We have seen how to characterise the relationship between variables with correlation and linear regression. We also said that **correlation is not causation**: an observed correlation does not necessarily imply a cause-effect relationship. Yet in many cases it is useful — or required — to establish one:

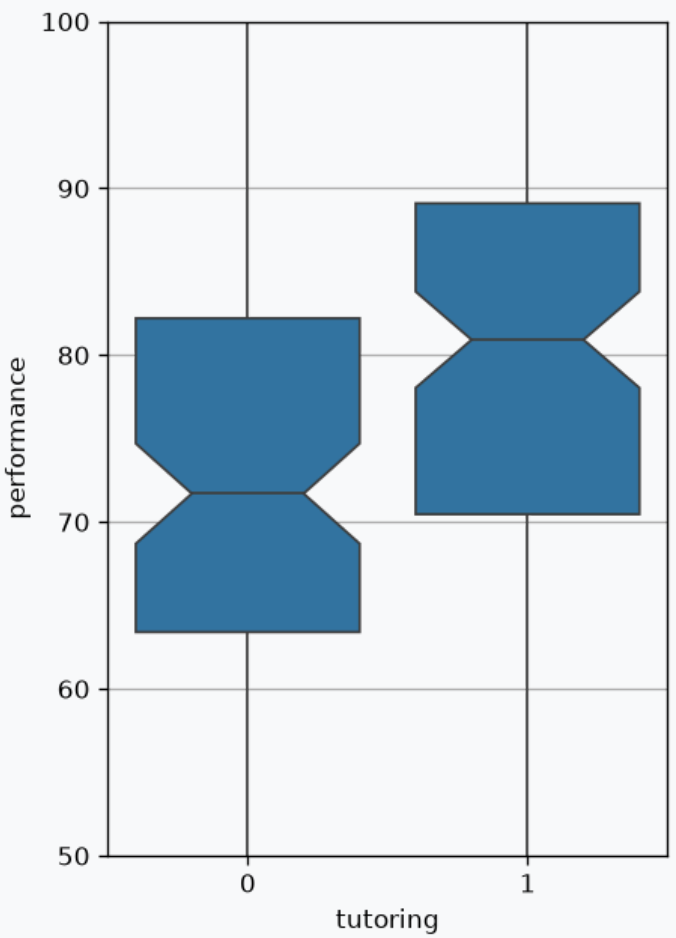
- We assess that **a given disease is more frequent among smokers**. Just a correlation, or a cause-effect relationship? If a smoker stops smoking, will their risk be reduced?
- We develop **a new drug** and assess that people taking it are less likely to get disease X. People should take the drug only if the benefits are clear.
- A large company finds that **sites close to the sea are more productive** than the others. Shall we move all sites closer to the sea?

- The field of **causal analysis** has developed a set of tools to answer these questions. This lecture is a primer: the main concepts, and some techniques that apply in simple cases.

Tutoring vs Performance

A school wants to see if a new teaching method is effective. They ask students if they want to **voluntarily enroll in a tutoring program**. After one year, they compare the average performance of students who participated with those who did not.





The comparison tells us `tutoring` seems to **significantly improve** performance.

We took a statistics class, so we check. The **notches** mark the confidence intervals of the medians: they do not overlap.

Since the notches do not intersect, the 95% confidence intervals do not overlap, so the two medians are unlikely to be close. To be very sure, run a **two-sample t-test**: the null hypothesis is that the two samples have the same mean.

```
T-statistic: 4.087
p-value: 6.36e-05
```

It looks like the tutoring program is a great success. **Is it?**

Measuring Causation: Potential Outcomes

The comparison seems clear: enroll everybody in the tutoring program! But **correlation is not (necessarily) causation**. We are seeing a correlation — is it also causation?

To assess a causal relationship we would ideally **take the same student and let them attend and not attend the tutoring at the same time**, and compare the effect of tutoring alone. This is impossible — and unethical: can we ask a student not to follow a programme we suspect will help? Even asking the same student to follow the tutoring and then stop does not work: the student "before" and "after" is not the same student, so the two values are never fully comparable.

We enter the realm of **potential outcomes**. We ask questions such as **"what if this student did not take the tutoring?"**

The notation

- T_i: the **treatment**, 1 if student i attended the tutoring, 0 otherwise. This is the standard terminology of causal inference, borrowed from the health sciences. The observations with $T = 1$ are the **treatment group**, those with $T = 0$ the **control group**.
- Y_i: the **outcome**, here the performance of student i.
- $Y_i(t)$: the outcome of unit i when the treatment takes value t.

Potential outcomes

For a given unit we **either observe $Y_i(0)$ or $Y_i(1)$, never both**.

The **potential outcome** is the value we *would* observe if we could intervene on T_i and change it. If $T_i = 0$ we observe $Y_i(0)$ but we also **theoretically define** $Y_i(1)$: it is not real and cannot be observed, yet it is a useful theoretical object.

Potential outcomes are also called **counterfactuals**, because they answer "what if...".

Treatment Effects: ATE and ATT

If we had potential outcomes, for each unit we could compute the **individual treatment effect** — the effect of attending the tutoring:

$$Y_i(1) - Y_i(0)$$

Averaging it over the distribution of observations gives the **average treatment effect**:

$$ATE = E[Y(1) - Y(0)]$$

It tells us **how performance varies on average if the same subjects take the tutoring or not** — the **causal effect** of tutoring on performance.

Alternatively, we can restrict the computation to the treated subjects, obtaining the **average treatment effect on the treated**:

$$ATT = E[Y(1) - Y(0) \,|\, T = 1]$$

which **also establishes a causal effect**.

- Both quantities need $Y(0)$ **and** $Y(1)$ for the same units, so neither can be computed from data. What can we compute instead?

What We Can Measure

Without potential outcomes we cannot compute ATE or ATT. We have **two distinct groups**: the **treated** (units with $T_i = 1$) and the **control group** (units with $T_i = 0$).

We could replace the calculation with the only data we can observe:

$$E[Y(1)|T = 1] - E[Y(0)|T = 0]$$

This is what **a correlation can measure**: the change in the average of Y across two distinct groups with different characteristics.

There are no potential outcomes in this expression: we directly observe $Y_i(1)$ when $T_i = 1$ and $Y_i(0)$ when $T_i = 0$. But it is now **comparing different groups**. The subjects who took the treatment are different from the ones who did not, and we do not know if their scores are really comparable.

> Imagine that, for some reason, students in the treated group are, by chance, better students. Then the difference we measure is biased and does not reveal a cause-effect relationship.

Let us show this in formulas.

Deriving the Bias

Start from what we measure, and **add and subtract the counterfactual** $E[Y(0)|T=1]$:

$$E[Y(1)|T=1] - E[Y(0)|T=0] = E[Y(1)|T=1] - E[Y(0)|T=0] - E[Y(0)|T=1] + E[Y(0)|T=1]$$

We note that, by linearity of the expectation:

$$ATT = E[Y(1) - Y(0)|T=1] = E[Y(1)|T=1] - E[Y(0)|T=1]$$

Regrouping the terms, the expression above becomes:

$$\begin{aligned} E[Y(1)|T=1] - E[Y(0)|T=0] &= \underbrace{E[Y(1)|T=1] - E[Y(0)|T=1]}_{ATT} + E[Y(0)|T=1] - E[Y(0)|T=0] \\ &= ATT + E[Y(0)|T=1] - E[Y(0)|T=0] \end{aligned}$$

We call the last term the **bias**:

$$BIAS = E[Y(0)|T=1] - E[Y(0)|T=0]$$

The Result: Measured Effect = ATT + BIAS

$$\underbrace{E[Y(1)|T=1] - E[Y(0)|T=0]}_{\text{what we measure (correlation)}} = \underbrace{ATT}_{\text{what we want}} + \underbrace{E[Y(0)|T=1] - E[Y(0)|T=0]}_{BIAS}$$

The effect given by the correlation (left term) is **biased**. The bias cannot be computed — it contains a counterfactual term, $E[Y(0)|T=1]$ — but we can interpret it:

> The bias measures how the treated ($T=1$) and control ($T=0$) groups would differ if no one had taken the treatment.

Reading the Bias

Similar groups

Treated and control are mixed students with similar abilities. Without the treatment they would perform the same on average, so the **bias is small or zero**.

Dissimilar groups

Control students are all better students. Even without anyone taking the treatment they would perform better on average — which **has nothing to do with the tutoring** or with cause-effect relationships. The **bias is different from zero**.

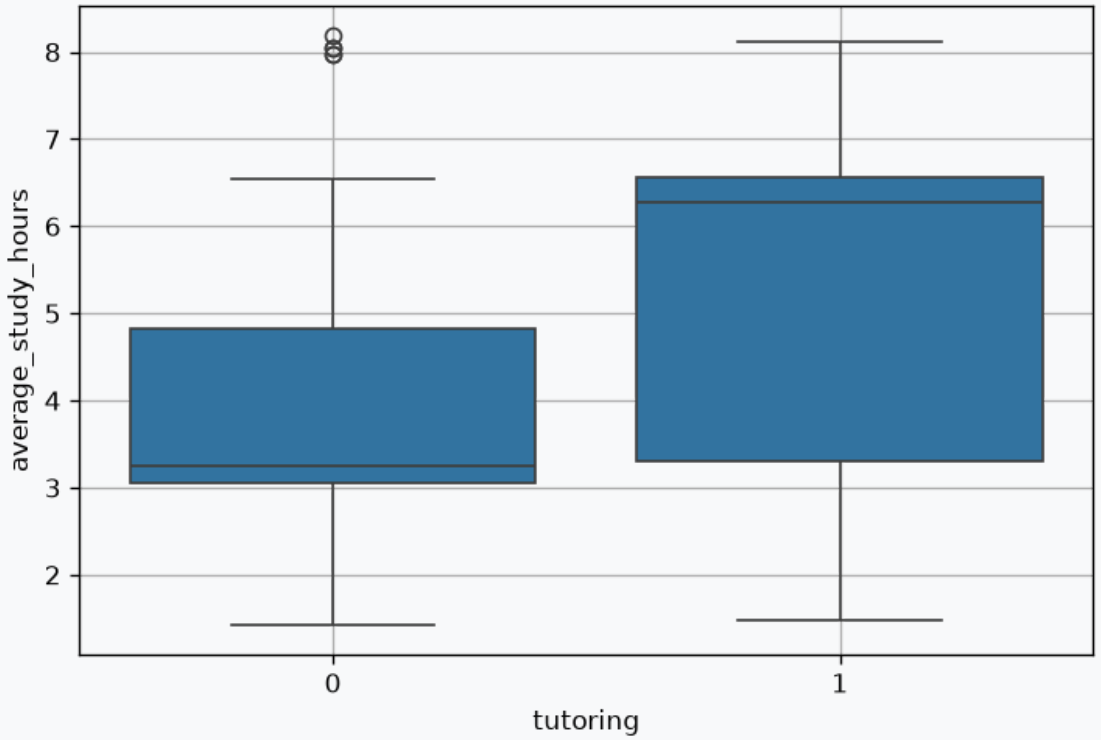
We now know **when association allows us to establish a cause-effect mechanism**: when the bias is zero —

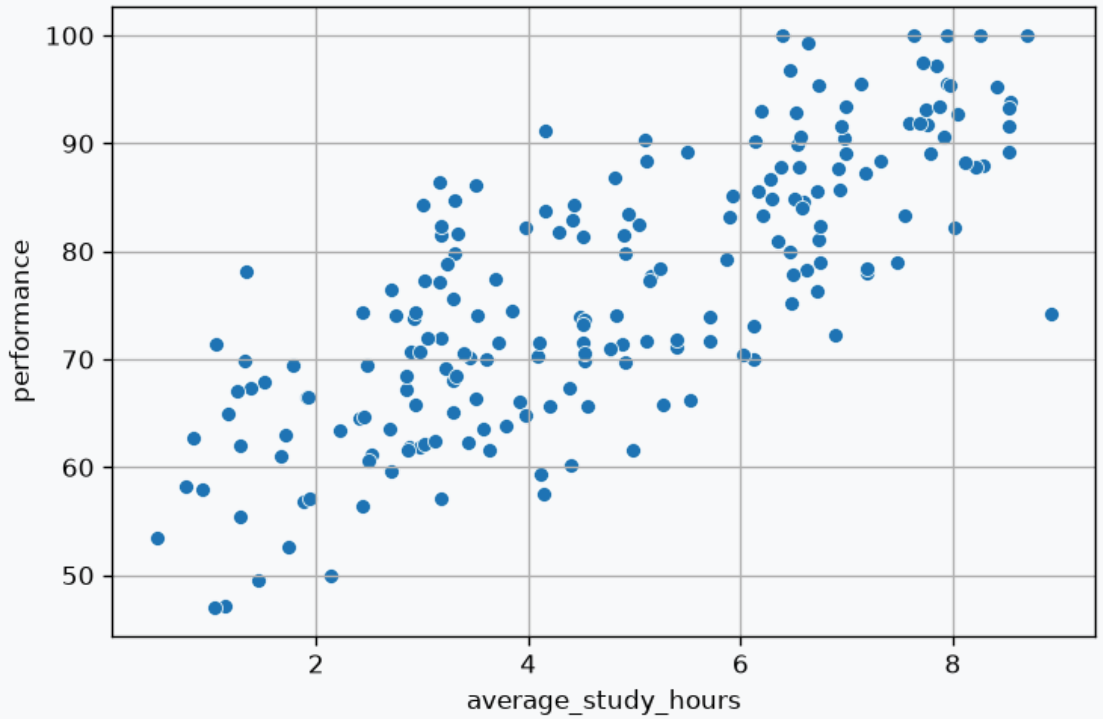
$$BIAS = E[Y(0)|T = 1] - E[Y(0)|T = 0] = 0$$

that is, when the difference between treated and untreated, **had no one been treated**, is zero (or at least very small). Then $E[Y(1)|T = 1] - E[Y(0)|T = 0] = ATT$.

Is the Bias Zero Here?

We cannot measure the bias — we have no counterfactuals. But we can look for a proxy of student ability: we ask how many hours per day they study, and check whether it is equally distributed in the two groups.





Students' skills are **not equally distributed** in the two groups (left), and study hours seem **correlated with performance** (right).

This is not enough to say the bias is non-zero — we do not observe counterfactuals — but the difference in outcomes could now have **two different causes**:

- Tutoring improves performance.
- Students following the tutoring study more on average (**maybe they are more motivated, so they study more and also choose the tutoring program**), and this affects the final performance.

Randomized Controlled Trials

We have seen that correlation is causation when treated and control groups are comparable in everything except the treatment. The most robust way to remove the bias is a **randomized experiment**, or **Randomized Controlled Trial (RCT)**.

A randomized experiment **randomly assigns individuals to the treatment or control group**. Of course, one of the two groups takes the treatment, which will likely make the groups distinguishable — but we want to be sure that, **apart from the treatment, the two groups are indistinguishable**.

It can be shown that random assignment makes the potential outcomes conditionally independent given the treatment:

$$Y(0) \perp Y(1) \mid T$$

If T is fixed, the potential outcomes are independent. For $T = 1$ (students who attended), knowing the observed $Y_i(1)$ does not tell us anything about $Y_i(0)$ — the performance the same student would have had without the tutoring.

Recall height and vocabulary, conditionally independent given age: they are independent within the same age group but not across groups, so **age is the only thing that makes them dependent**. Likewise, if $Y(0) \perp Y(1)|T$, then **the treatment is the only thing generating a difference** between treated and control.

From Randomization to Causation

If the conditional independence holds, the expected untreated outcome does not depend on the group:

$$E[Y(0)|T=0] = E[Y(0)|T=1] = E[Y(0)]$$

and the same holds for the treated outcome:

$$E[Y(1)|T=0] = E[Y(1)|T=1] = E[Y(1)]$$

Substituting in the measured difference:

$$E[Y(1)|T=1] - E[Y(0)|T=0] = E[Y(1)] - E[Y(0)] = E[Y(1) - Y(0)] = ATE$$

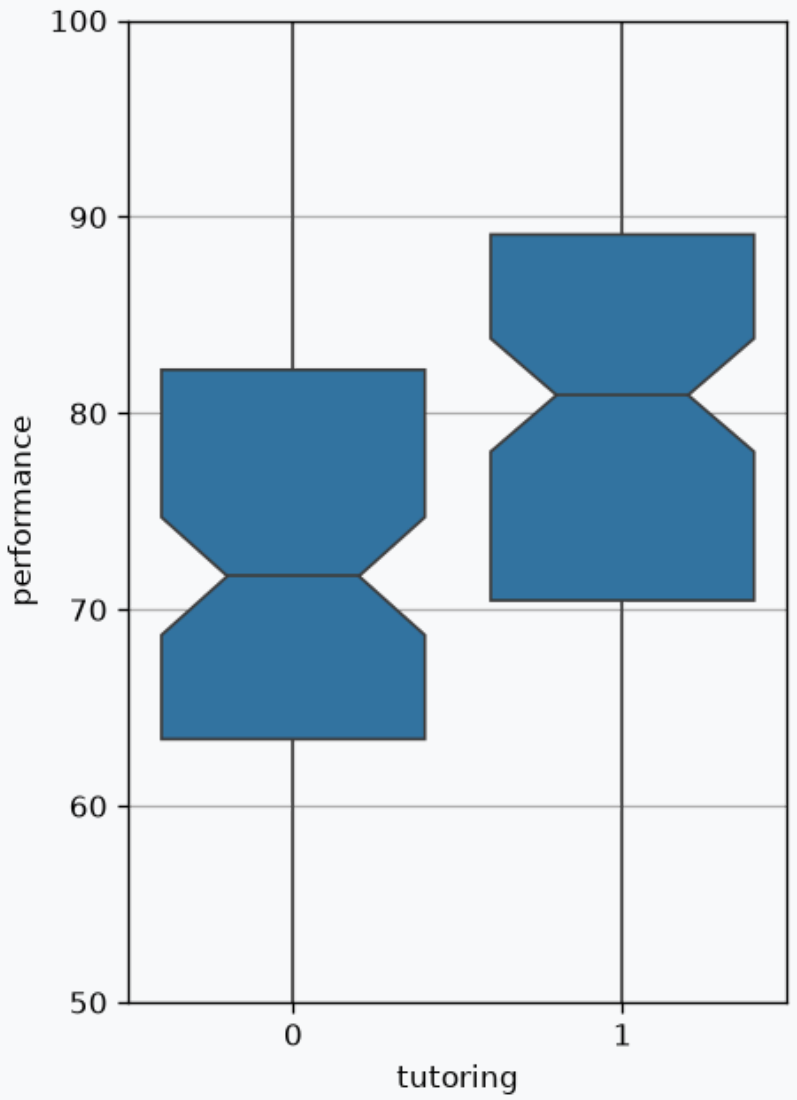
In terms of the decomposition we derived: the bias $E[Y(0)|T=1] - E[Y(0)|T=0]$ is zero, and $ATT = ATE$.

In this case, **correlation is causation**.

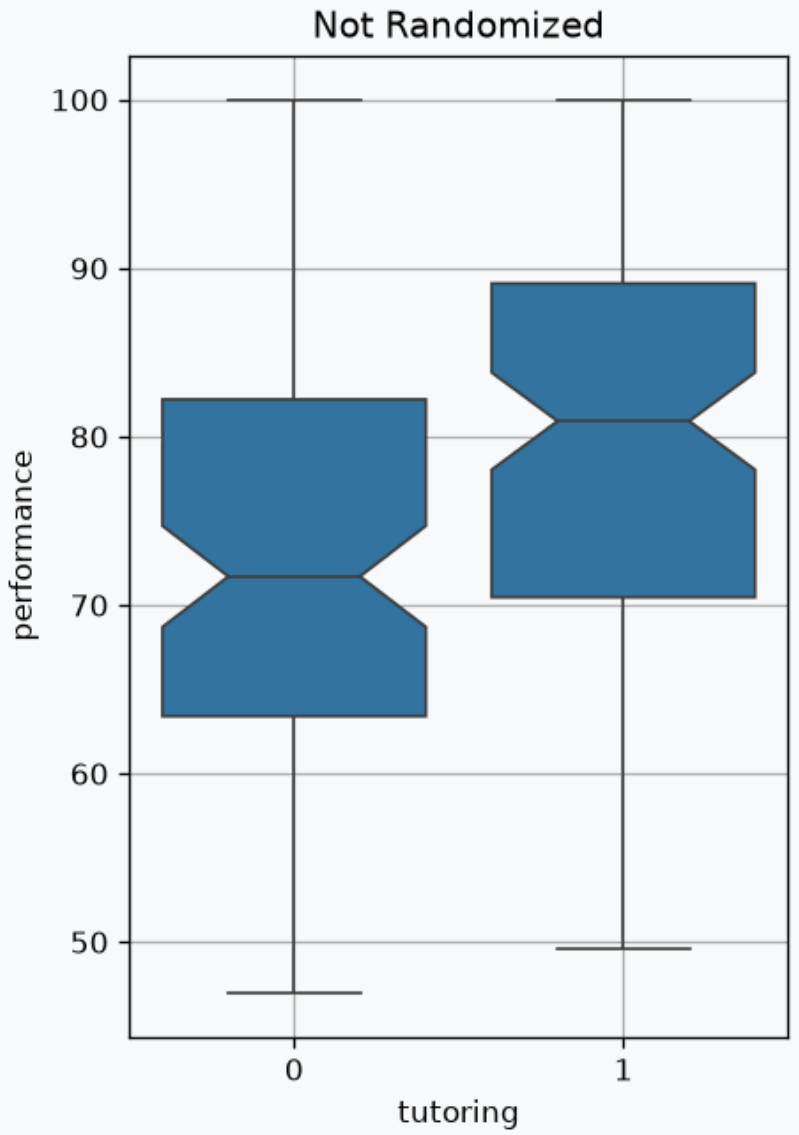
Randomizing Tutoring

Let's repeat the experiment with random assignment. (A word of caution: this is not always possible — it would not be very polite to "force" students to enroll in a programme, but let us assume we can.)

Not randomized



Randomized



We now see a **less dramatic, and actually negative** difference. What happened?

By randomising we got rid of **selection bias**. Students who volunteered were probably more motivated and keen to study; the effect of tutoring itself is very small. A t-test confirms it:

```
T-statistic: -1.133
p-value: 0.259
```

The p-value is large: the difference is **not statistically significant**.

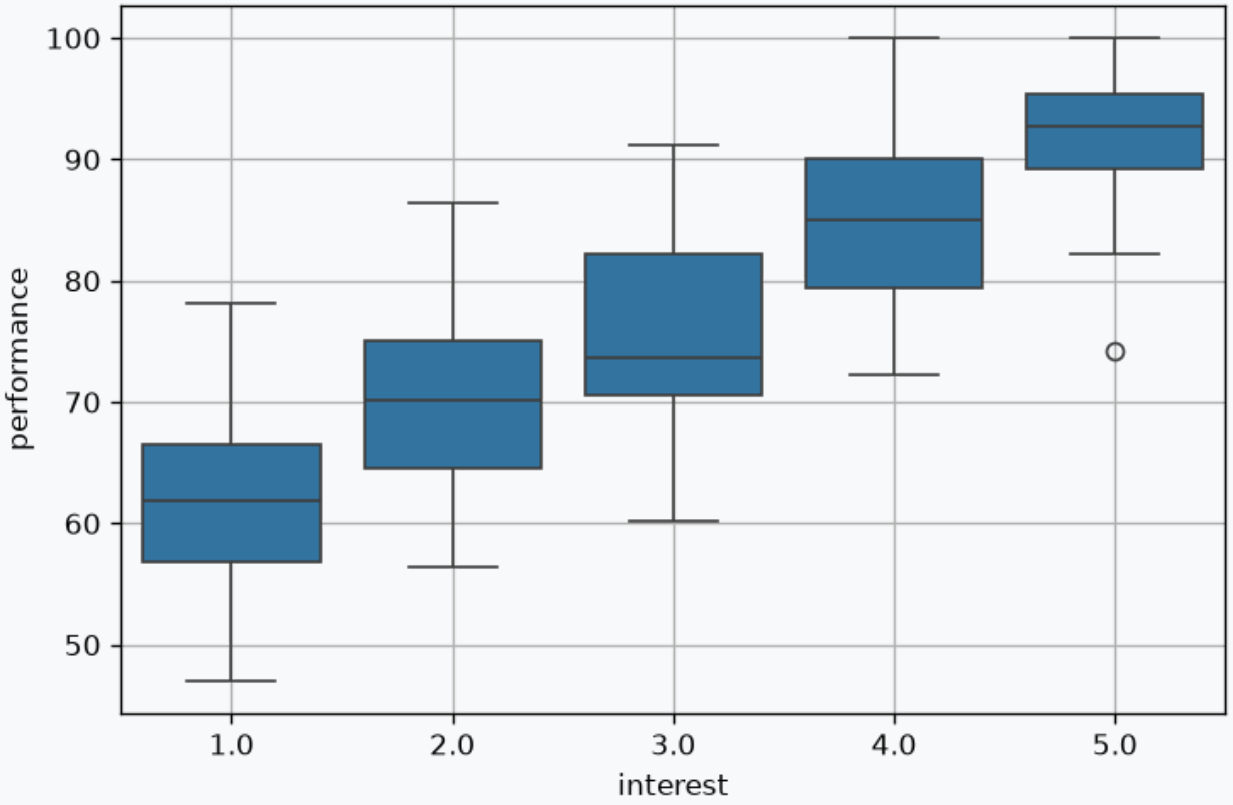
Causal Effect and Observational Studies

The gold standard is the randomized controlled trial, but **it is not always feasible or ethical**. Consider estimating the effect of smoking on the development of a disease: we would have to select subjects randomly in two groups and ask one group to smoke — **unethical**, since we suspect smoking is bad for health.

In these cases we can only resort to **observational data**: we collect data on smokers and non-smokers and observe the association. We know this process is **subject to bias**. How do we deal with it?

- **Graphical causal models** give us a framework to **set our beliefs on where the bias comes from**. If we can model the source of bias accurately, we can remove it by **controlling** on the variable "causing" the bias.

Back to the students: assume we cannot randomise the assignment (again, it is unethical). Since we suspect that **the interest of students towards study is a source of bias**, we also ask them how interested they are in studying, on a scale from 1 to 5.

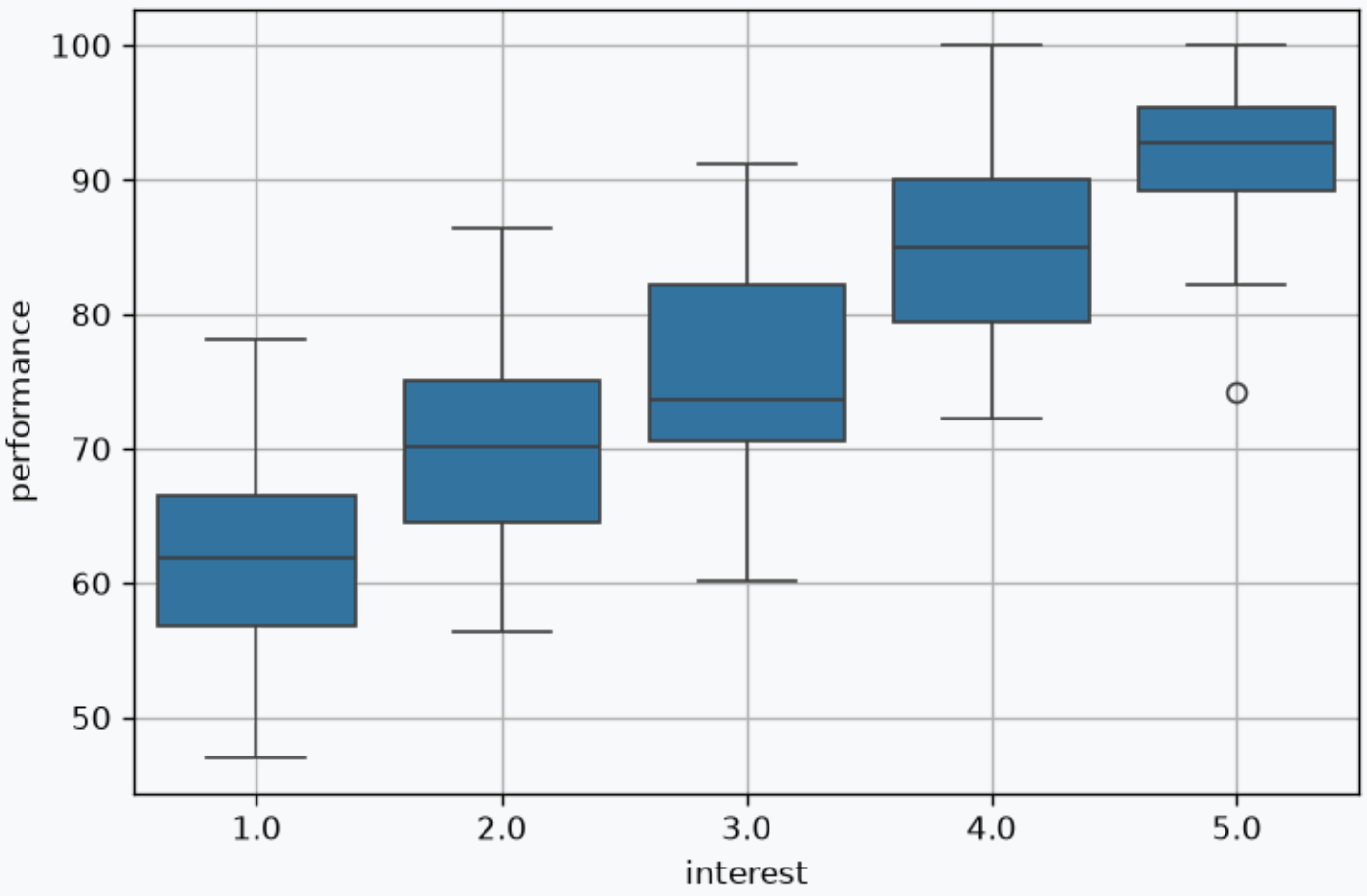


These are **the same observations as the first experiment**, with one variable added: `interest`.

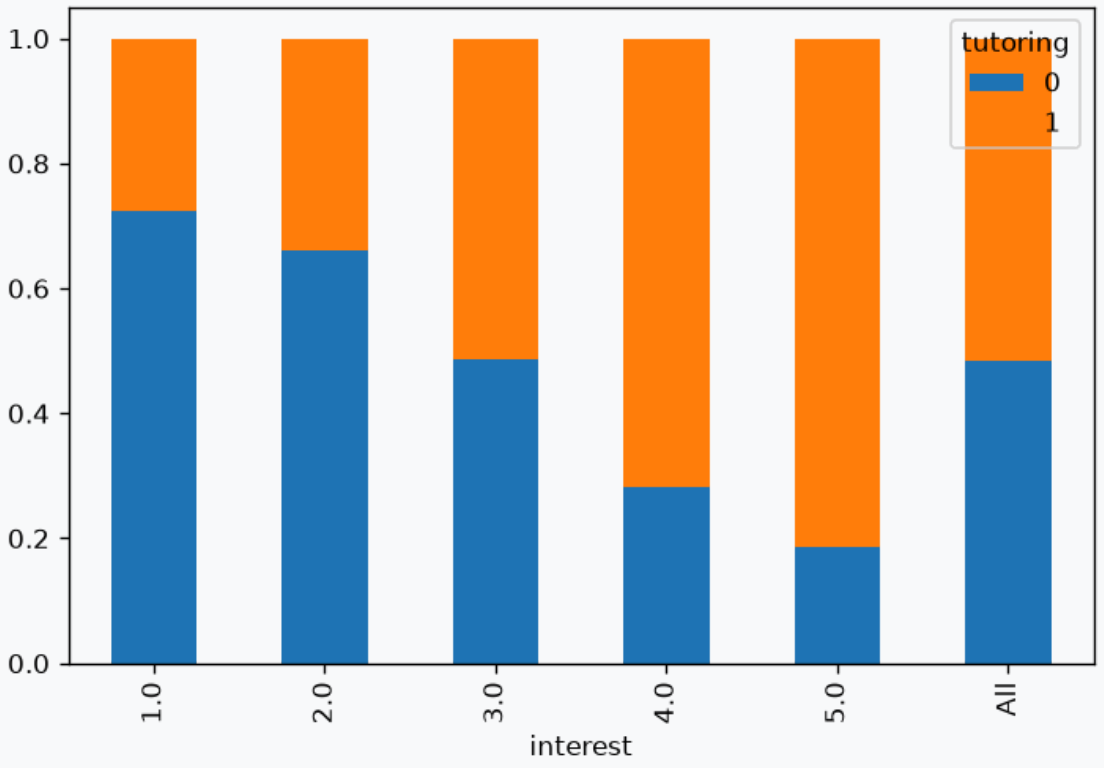
Is `interest` Really the Source of Bias?

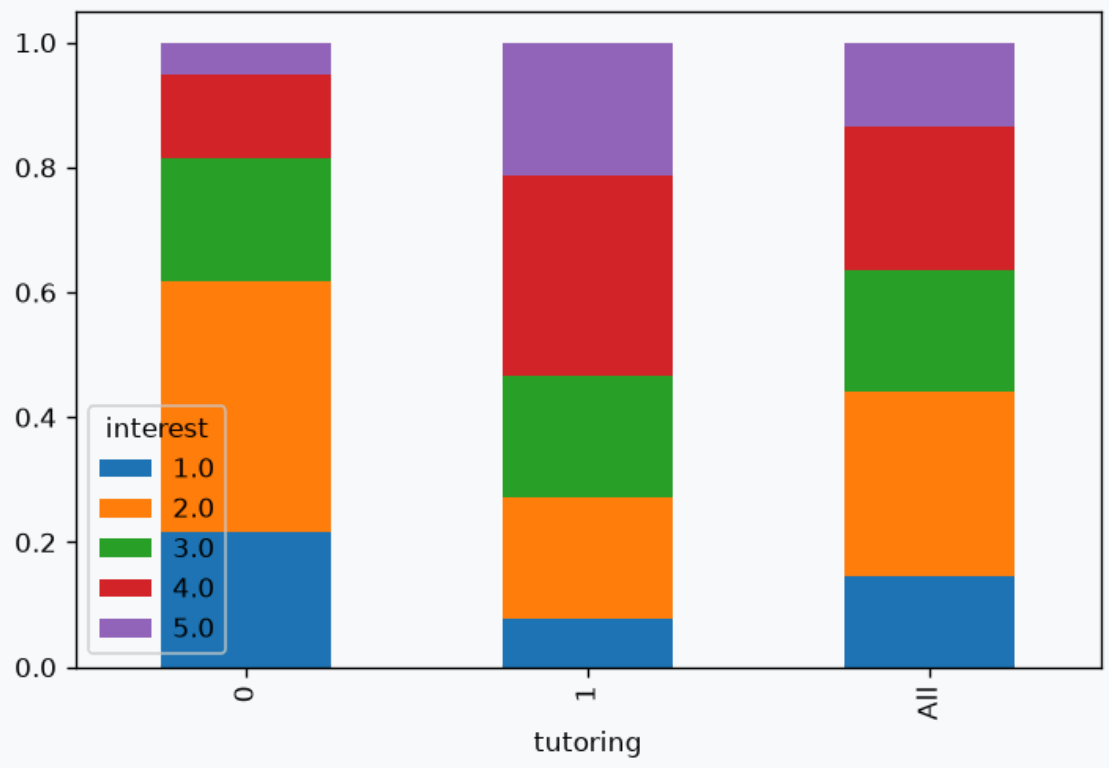
We suspect that **interested students perform better** and that **interested students are more likely to enroll** in the tutoring program. Let us check.

Performance split by interest — we were right! (an exaggerated picture, the example is made up):



Share of treated and untreated within each interest level, and the distribution of interest in the two groups:





Is `interest` Really the Source of Bias? — The Test

A χ^2 test of independence tells us whether the association between `interest` and `tutoring` is statistically significant:

```
Chi-square statistic: 31.22
Cramer V statistic: 0.40
Chi-square p-value: 2.76e-06
```

We were right in this case too: **interest affects tutoring**, hence interest is our source of bias.

- The correlation between `interest` and `tutoring` makes the treated (`tutoring=1`) and control (`tutoring=0`) groups **not comparable**. If people in the treated group have more chances of getting a higher performance, no wonder we see such dramatic differences!

  The effect we observe may be due to the bias introduced by `interest` , to the effect of `tutoring` , or to a **combination of both**.

Controlling for `interest`: the Idea

How do we make the groups more comparable? Recall that we want:

$$Y(0) \perp Y(1) \mid T$$

We have seen that this is **not true** because of `interest`. If we pick a person in the group $T = 0$ with a high $Y(0)$, we can imagine this is due to a large `interest`, so we can predict that $Y(1)$ will also be large.

> If we can predict $Y(1)$ from $Y(0)$, then they are **not independent**.

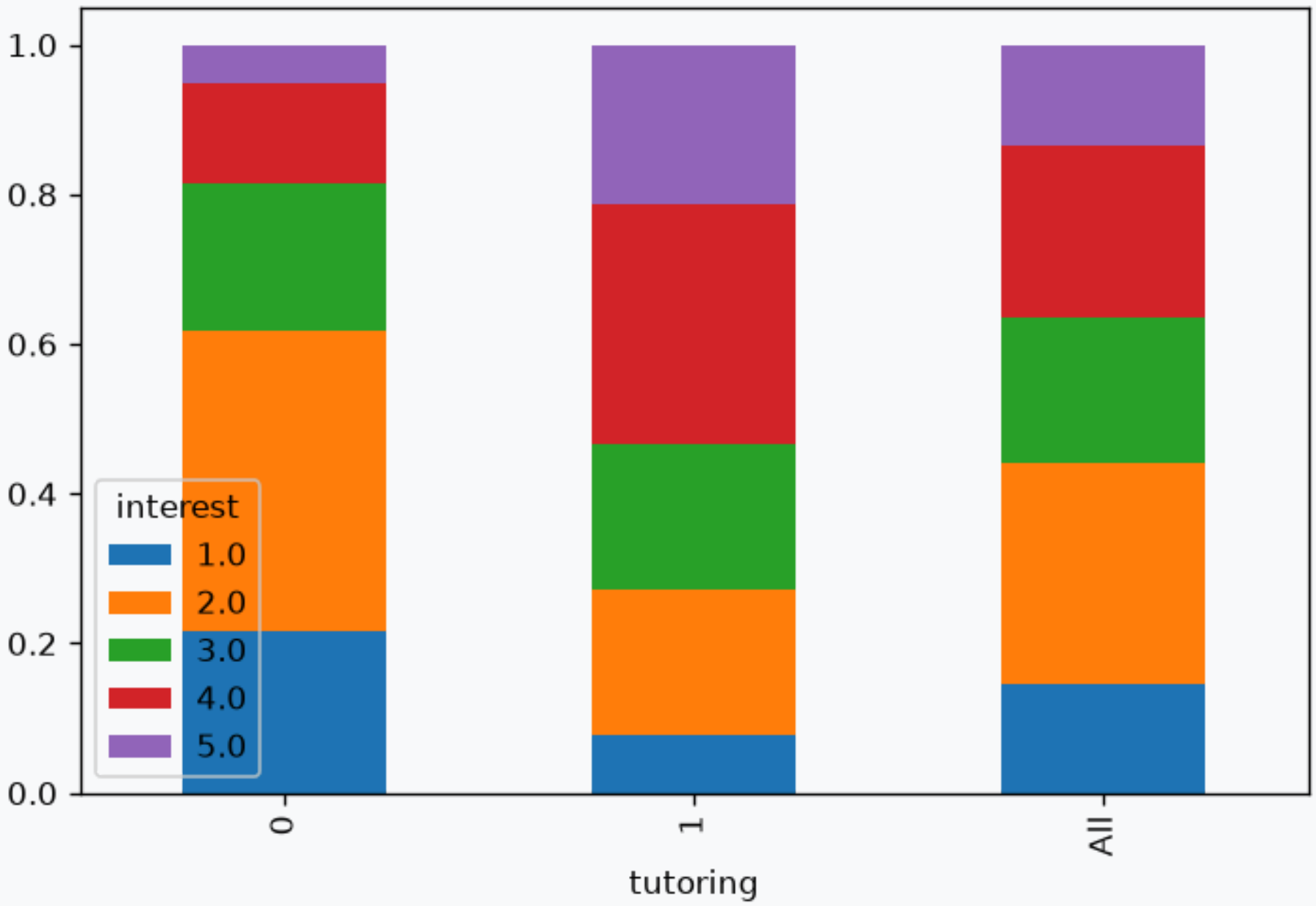
Obtaining the statement would require **random assignment**, which we cannot do. However, we know `interest` is the source of bias. Let us assume it is the **only** source of bias. Then, if X represents `interest`:

$$Y(0) \perp Y(1) \mid T, X$$

Why? Pick a person from the group $T = 0$ with a given interest, say $X = 4$, and a large $Y(0) = 80$: a student with great interest and high performance. What can we say about $Y(1)$? Perhaps that it will be high, because the student has high interest — but any other student in this group will probably have a high score too, so we cannot predict anything significantly different from the score of another random student **within this group**.

Controlling for `interest`: the Data

The easiest way to condition on X is to **divide the data into groups based on the values of X and assess any association within the groups**.



Within each `interest` group the association between `tutoring` and `performance` tends to **vanish**: the notches (confidence intervals) tend to intersect. A clearer picture comes from **5 two-sample tests**, one per group, on the means of `performance`:

interest	t statistic	p-value
1	0.935	0.358
2	−0.781	0.438
3	−0.867	0.391
4	0.165	0.870
5	−1.383	0.179

No significant association between `performance` and `tutoring`: this suggests **the lack of a causal effect**.

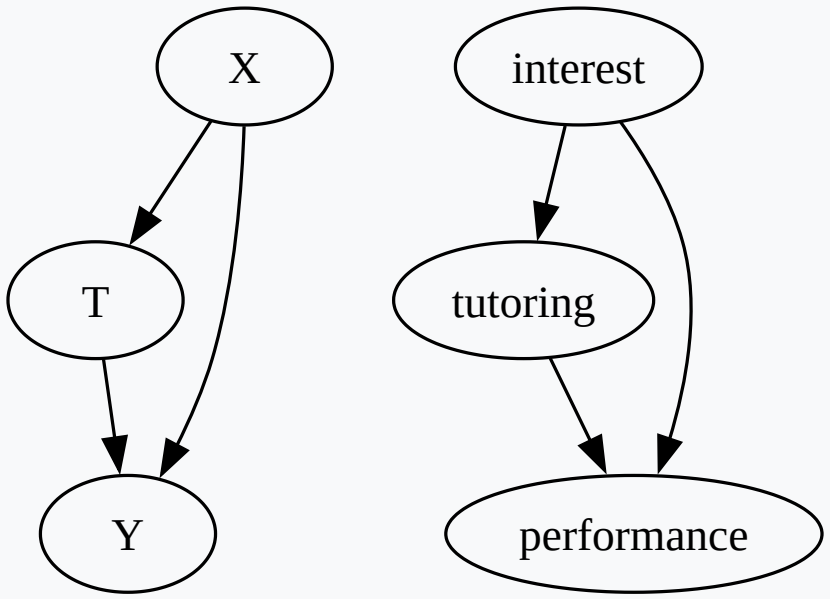
- The act of conditioning on X (for instance by dividing into groups) is called **"controlling for X"**. But how do we know **which** variables to control for? Controlling for the wrong one makes things worse.

Graphical Causal Models

We have seen a simple case in which we could identify the main source of bias. Not all cases are so simple: **more than one variable may be responsible for bias**.

How do we identify the variables we should control for?

A **graphical causal model** is a **directed acyclic graph** formalising cause-effect relationships: **each node is a random variable, and a directed edge $A \to B$ says that A causes B.**



The graph describes our hypotheses: we expect `interest` (X) to influence both `performance` (Y) and whether a subject belongs to the treated or control group, i.e. `tutoring` (T). From the graph we can identify `interest` as a source of bias.

As we will see, **a set of rules** tells us which variables we should and should not control for.

- Graphical models are **designed by the data analyst**, who encodes their own **beliefs** about how the world works. They come from a Bayesian view: first state the beliefs, then use a theoretical framework to draw conclusions from them.

Main Structures of Causal Graphs

Once we form a graphical model, we can identify a set of standard structures that tell us which variables to control for. These are the main ones; others may exist.

Chains

$A \to B \to C$

Forks

$A \leftarrow B \to C$

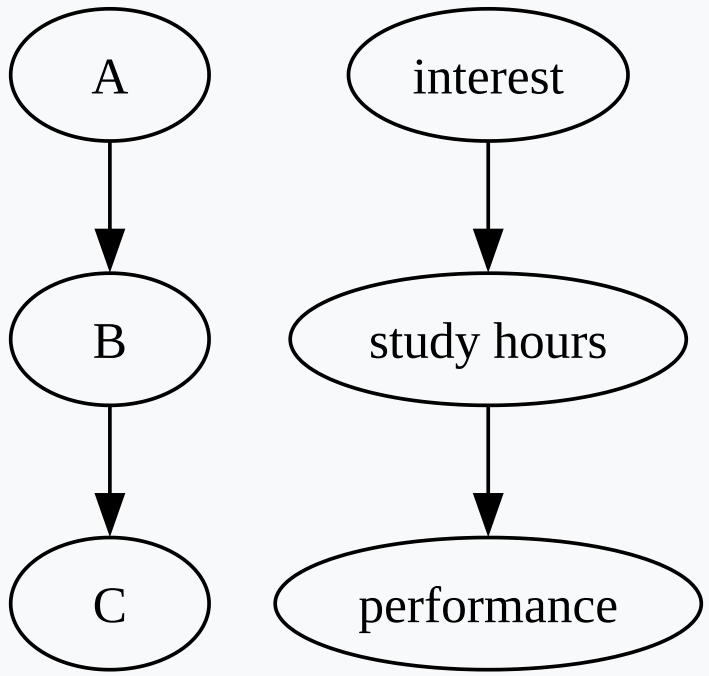
Colliders

$A \to B \leftarrow C$

For each we ask the same two questions: **are the end nodes dependent?** And **what happens when we condition on the middle node B?**

Chains

The most straightforward structure is a chain. `interest` causes `study hours`, which in turn causes `performance` ($A \to B \to C$):



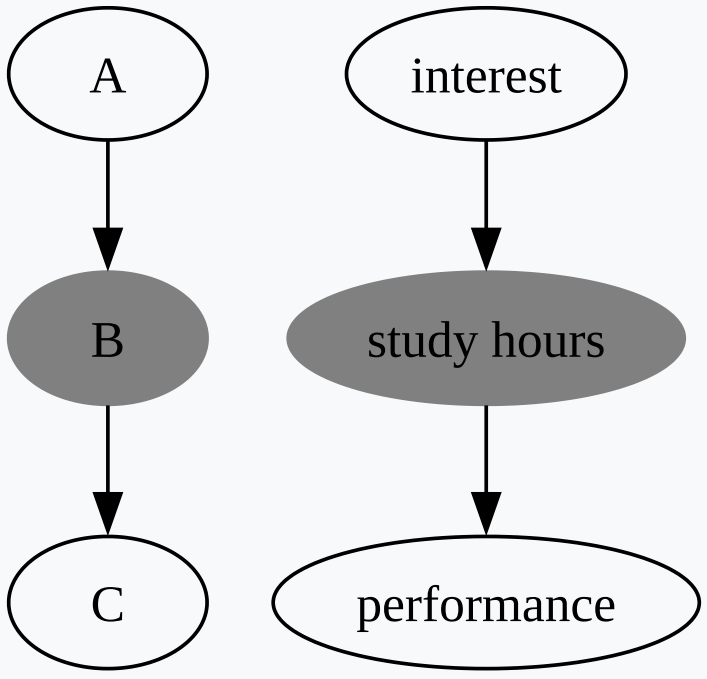
Let us imagine no other variable is involved: `interest` is the only cause of `study hours`, and `performance` is only caused by `study hours`. Then:

- `interest` and `study hours` are associated;
- `study hours` and `performance` are associated;
- `interest` and `performance` are associated.

Indeed: if a person has a great interest, they study a lot and have good performance; if a person has good performance, we infer they study a lot and hence have great interest; if a person studies a lot, we infer great interest and good performance.

Chains: Controlling for B

Let us **control for B,** `study hours`. We denote the variable we control for in gray:



If we control for B, `interest` **and** `performance` **are not associated anymore**:

> Consider a group of people who study the same number of hours, and take a person with good performance. We don't know if they have great interest or not: we cannot distinguish a person with great interest from one with little interest, as they all study the same amount.

In mathematical terms, in general:

$$A \not\perp C$$

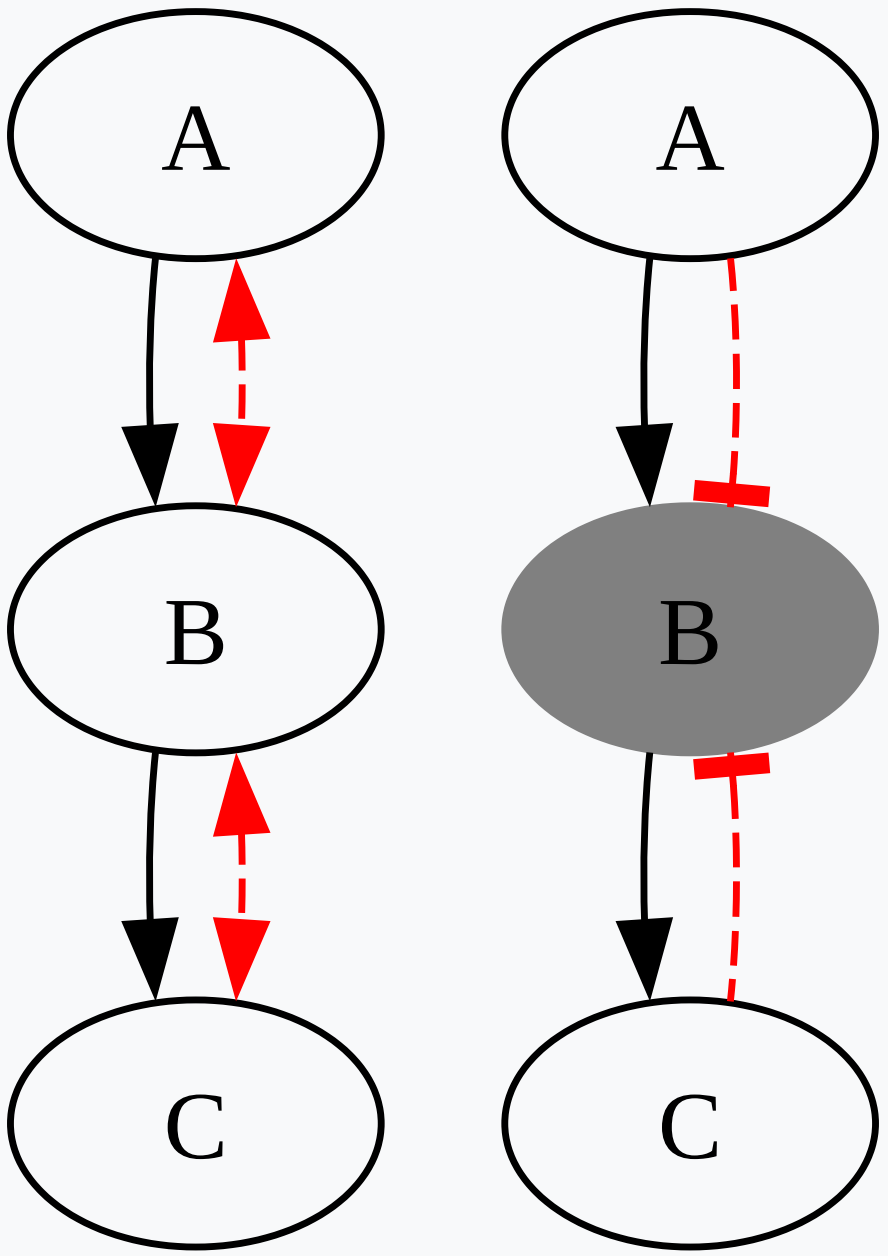
but, by controlling for B:

$$A \perp C \mid B$$

In groups with the same value of B, A and C are independent.

Chains: How Dependence Flows

Besides solid black arrows for cause-effect relationships, we also show **in red how dependency flows**. Two identical graphs; on the right we condition on B:



- **Left**: dependency flows in both directions **through** B. A depends on B and C, and C depends on B and A.
- **Right**: conditioning on B "stops" the dependency. A and C are independent of B — conditioning means fixing a value for B, so they are independent of a constant — and $A \perp C \mid B$.

Conditioning on B **blocks the dependency**: while there is a causal effect between A and C, **we cannot see it anymore**.

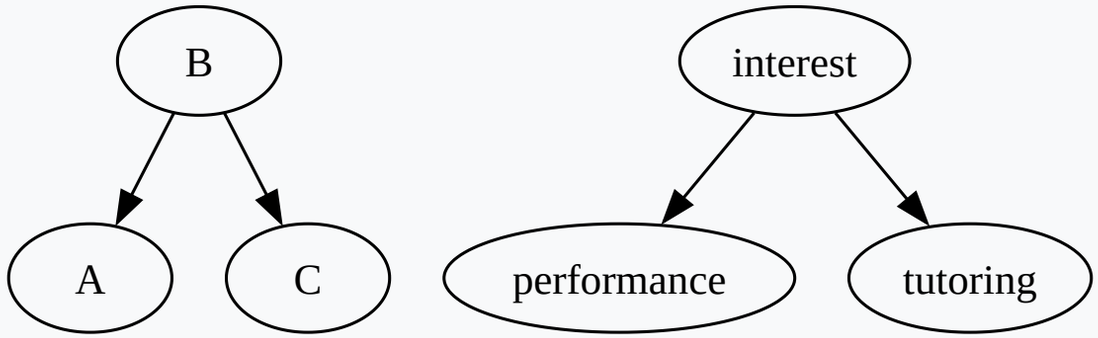
Chains: What Not to Do

If we want to study the relation between A and C in a chain, we **should not condition on B**: the association between A and C will vanish even though a causal effect exists.

- **Conditioning on any variable is not a good way to remove bias.** If we condition on the "wrong" variable, we may introduce bias that was not there in the first place, and estimate the lack of a causal effect even when the causal effect is there.

Forks

A fork is a variable that causes two other variables. B (`interest`) causes both A (`performance`) and C (`tutoring`) — the example we have seen before:



We will likely observe correlations between any pair of A, B, C. For instance a large value of `performance` probably means `interest` is high, hence `tutoring` will probably be 1. But the graph tells us there is **no cause-effect relationship between A and C**.

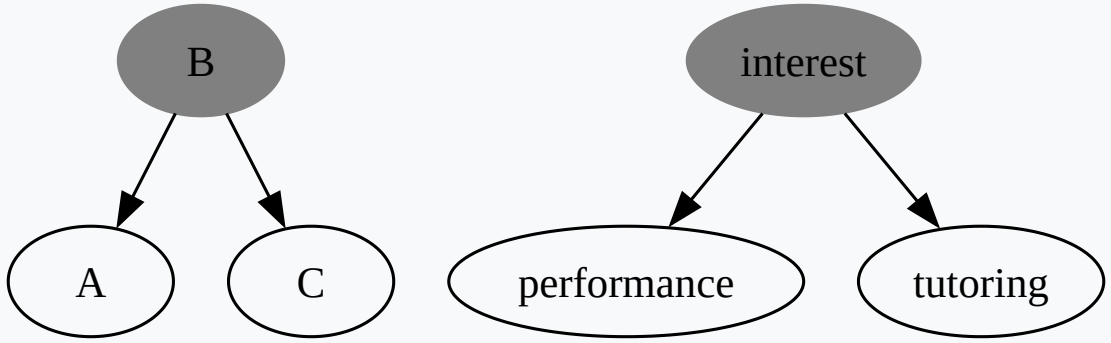
> We call B (`interest`) the **common cause**.

The fork structure **creates a backdoor path**: dependence (association) flows through B (the backdoor) to go from A to C.

Forks: Controlling for the Common Cause

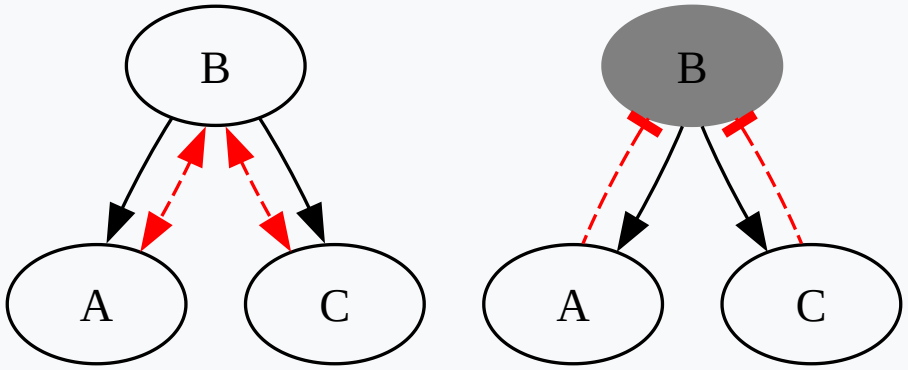
Conditioning on B (`interest`) makes the association between A and C vanish:

$$A \perp C \mid B$$



In this graph we observe **no association** between `performance` and `tutoring`, which confirms the lack of a causal effect — exactly what we saw when we compared students of the same interest.

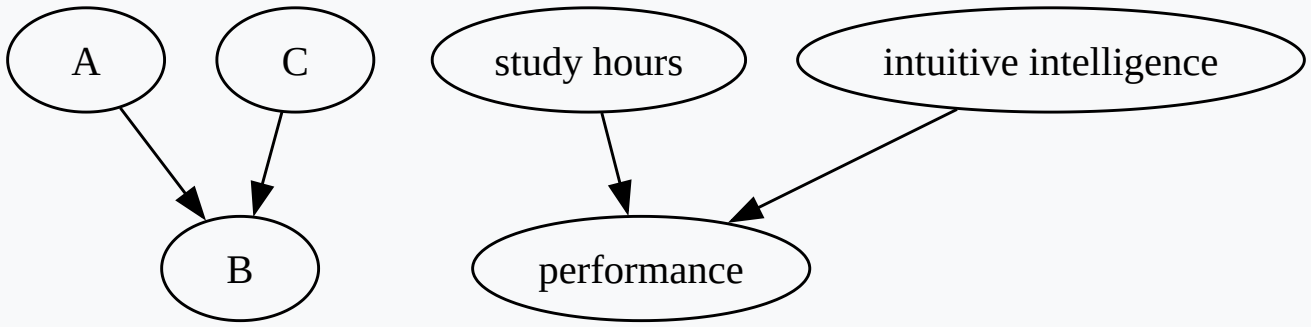
Forks: How Dependence Flows



- **Left**, B not conditioned: dependency (hence association) flows from A to C through B. **We observe an association which is not due to a causal effect.**
- **Right**, B conditioned: dependency does not flow through B, so A and C are independent and we measure no association, **as it should be**.

Colliders

The opposite of a fork: a variable is **caused by** two different variables. `performance` is caused by `study hours` and by `intuitive intelligence`:



We assume that `intuitive intelligence` does not affect `study hours` in any way. This might be inaccurate, but it keeps the example simple.

The structure is called a **collider** because two arrows (from A and C) collide on a single node (B).

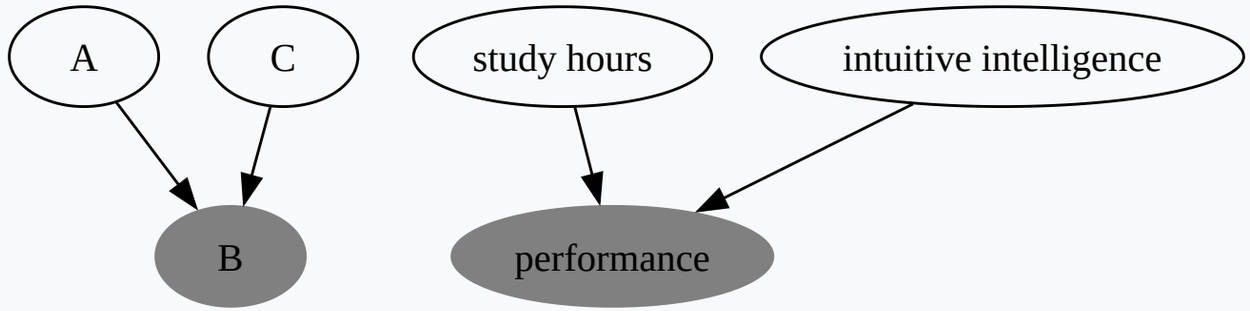
A collider **blocks dependency**. If we do not condition on `performance`, we observe no association between `study hours` and `intuitive intelligence`:

- if a person studies a lot, we cannot say anything about their intuitive intelligence;
- if a person has intuitive intelligence, we cannot say how much they study.

$$A \perp C$$

Colliders: Conditioning on the Collider

Let us now condition on B (`performance`):



Conditioning on `performance` **changes things**:

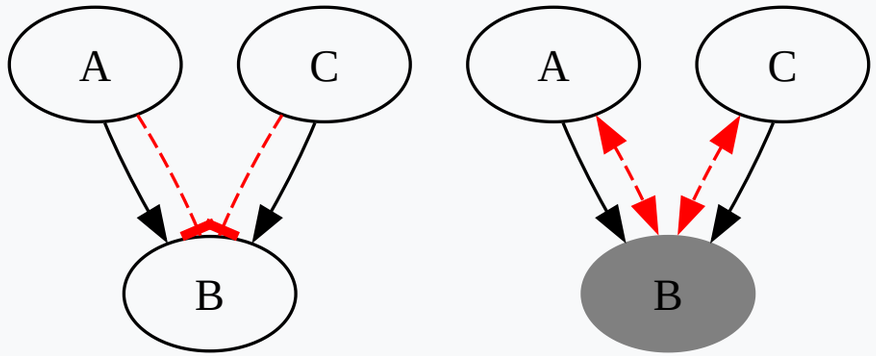
- Among students with **good performance**, knowing that one **does not study a lot** tells us they probably have good intuitive intelligence: if the performance cannot be explained by the many study hours, it must be due to intuitive intelligence.
- Conversely, if a good performing student has bad intuitive intelligence, their performance is probably due to a large number of study hours.

Conditioning on `performance` makes `study hours` and `intuitive intelligence` **dependent**:

$$A \not\perp C \mid B$$

Colliders: Explaining Away

This phenomenon is called **explaining away**: once we fix B (`performance`), knowing A (`study hours`) already explains B, which makes C less likely.



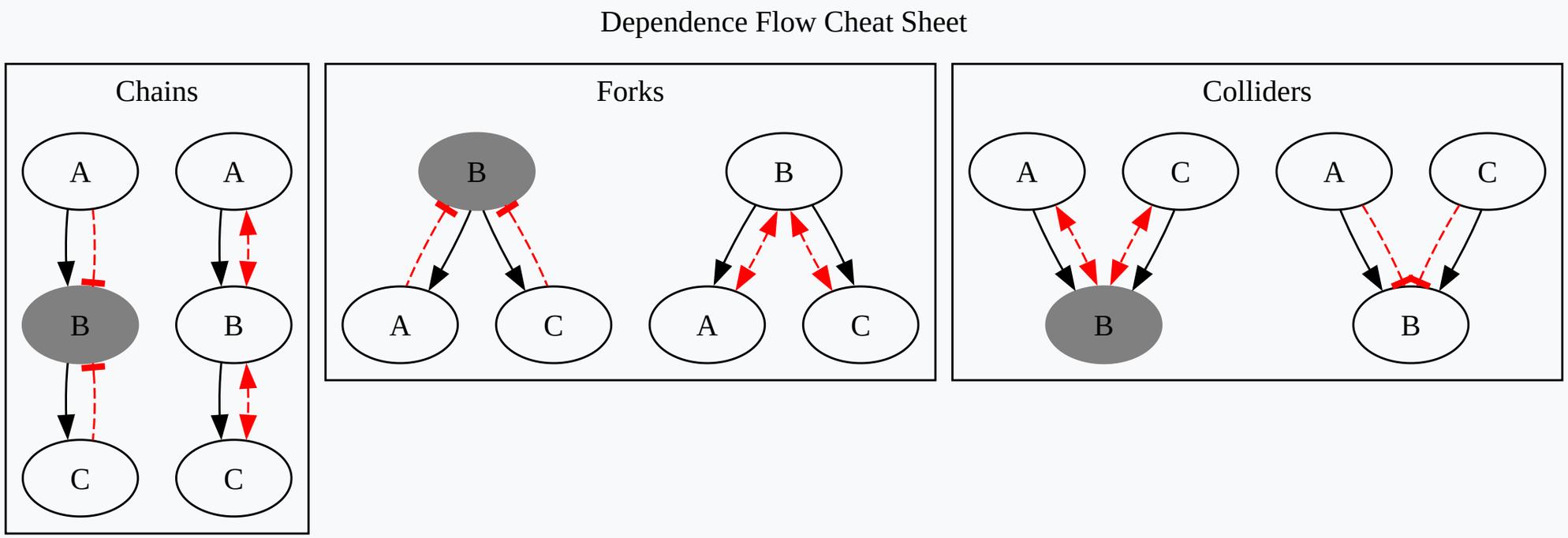
- **Left**, B not conditioned: the bar on the arrows says the flow is **blocked** at the collider.
- **Right**, B conditioned: dependency flows between A and C.

A collider **blocks dependency**, while conditioning on it **unblocks** a dependency path — and the association it creates is not a causal effect.

This is why controlling for a variable "just to be safe" is not safe.

Dependence Flow Rules

Two nodes are dependent when **an unblocked path exists between them**. Combining the three structures gives the cheat sheet:



Solid black arrows are cause-effect relations; dashed red arrows show how dependence flows. A grey node is a node we are conditioning on, and a bar across an arrow means the flow is blocked.

The cheat sheet is replicated from the Stanford graphical-models short course (lec. 2) and from Facure Alves' handbook, chapter 4.

Blocked Paths, and d-Separation

Two nodes are independent when **every path between them is blocked**. A path between A and B is blocked if:

It contains a non-collider that has been conditioned on

A chain or a fork through a grey node stops the flow.

It contains a collider that has not been conditioned on

...and none of whose descendants has been conditioned on either.

- Deciding whether two nodes are independent given a set of others is a **mechanical procedure** — trace every path, block it or not according to the three rules — and it is worth doing on paper a few times; the exercises at the end of the chapter ask you to.

  In practice you will rarely do it by hand: what matters for an analysis is **recognising confounding and selection bias**, which is what comes next.

Using Graphs to Diagnose Bias

We can now use graphical causal models to **diagnose bias**: which variables we should condition on, and which we should not, when measuring a causal effect through correlation.

There are two main sources of bias:

Confounding bias

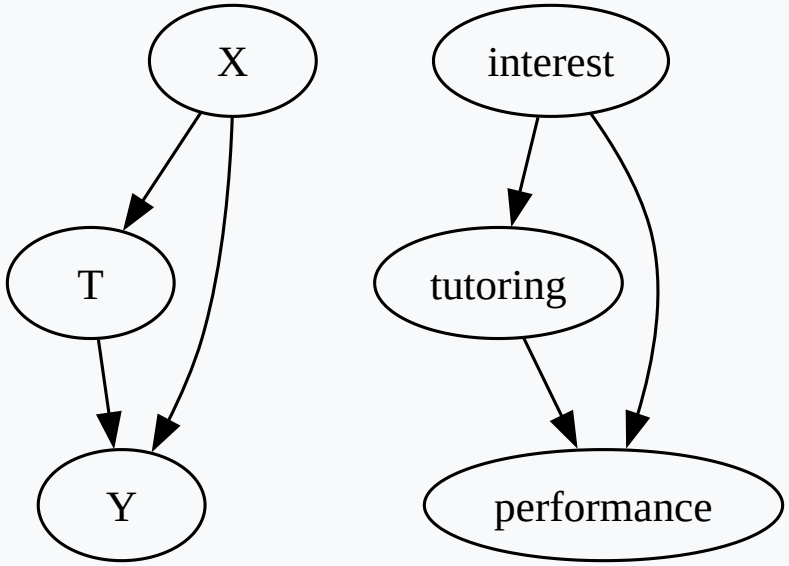
The treatment and the outcome have **a common cause** that we did not condition on.

Selection bias

We **control for too many variables**: a common effect, or a mediator.

Confounding Bias

Confounding bias happens when **the treatment and the outcome have a common cause**. Consider the example we already saw:

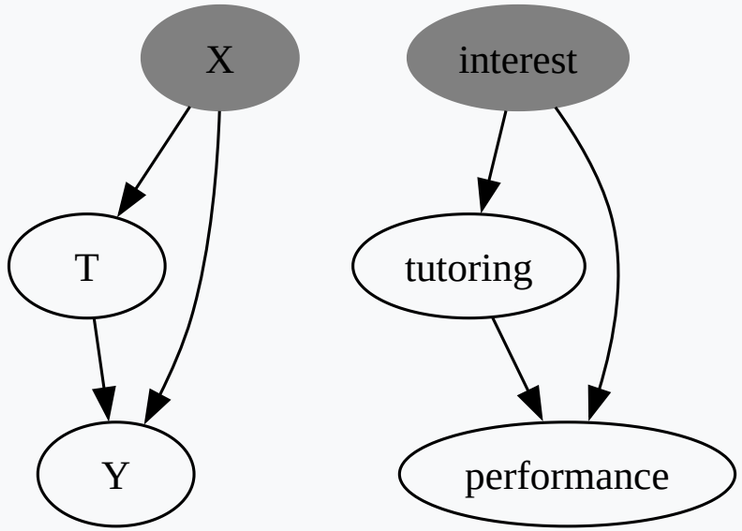


The outcome Y is `performance`, caused by both the treatment T (`tutoring`) and another common cause X (`interest`). X is a **confounder**: it mixes up the effect of `tutoring` on `performance` with the effect of `interest` on `performance`.

If we measure the correlation between `tutoring` and `performance` we get a large number, but we do not know whether it is due to the common cause: maybe people choose tutoring because they are interested, and obtain good performance because they are interested as well.

Confounding Bias: Closing the Backdoor

To measure the **direct effect between treatment and outcome** we need to **close the backdoor path between Y and T**, by **conditioning on the common cause**:



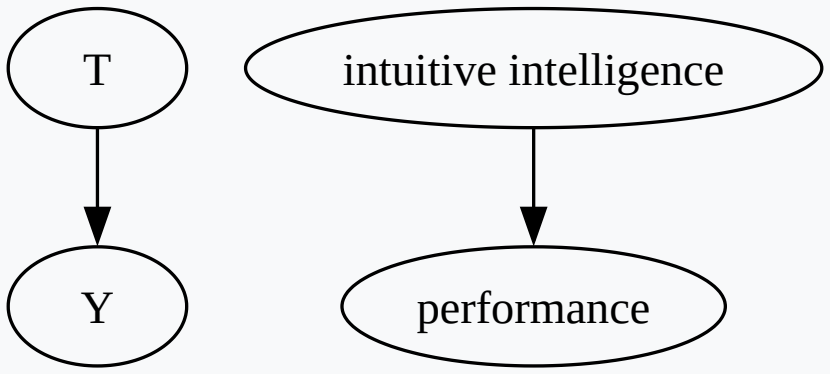
This corresponds to comparing subjects with the **same level of interest**. All students in the group have the same `interest`, so any association between `tutoring` and `performance` is given by the causal effect. By conditioning on X:

$$Y(0) \perp Y(1) \mid T, X$$

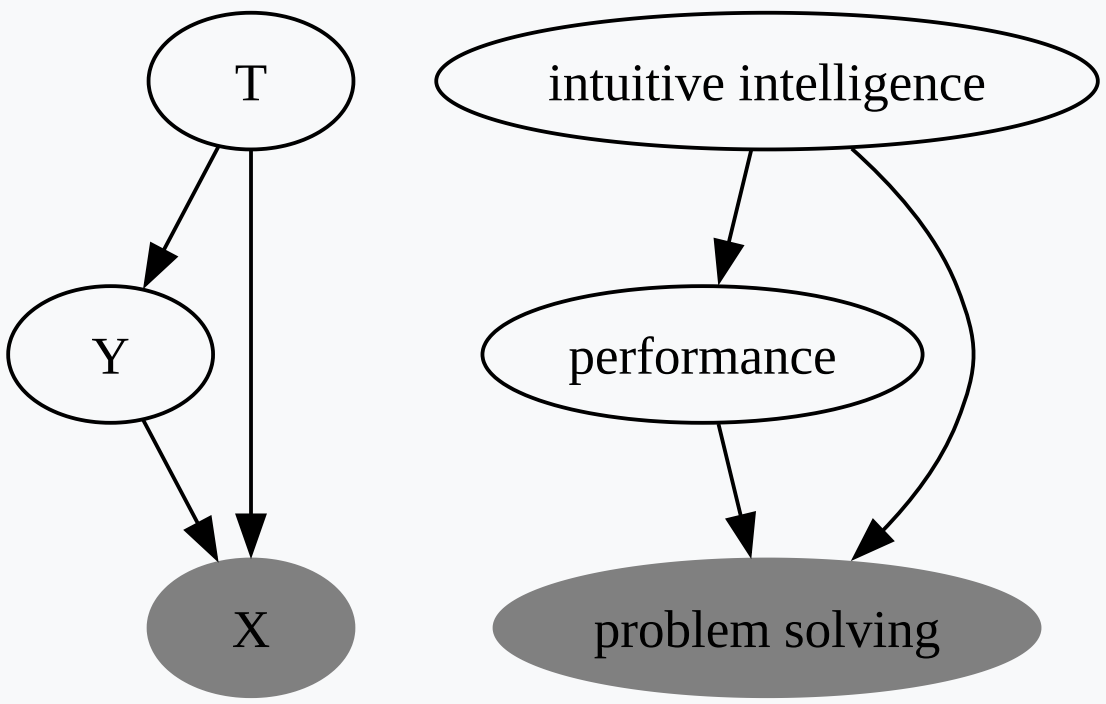
Selection Bias: Controlling for a Common Effect

While confounding comes from **not** conditioning on a common cause, selection bias comes from **controlling for too many variables**. One may think it is a good idea to control on any variable "just in case" — it is not.

We want the causal effect of `intuitive intelligence` on `performance`:



In our data we also have the score of a test on `problem solving`. **"Just to be sure", we condition on it:**



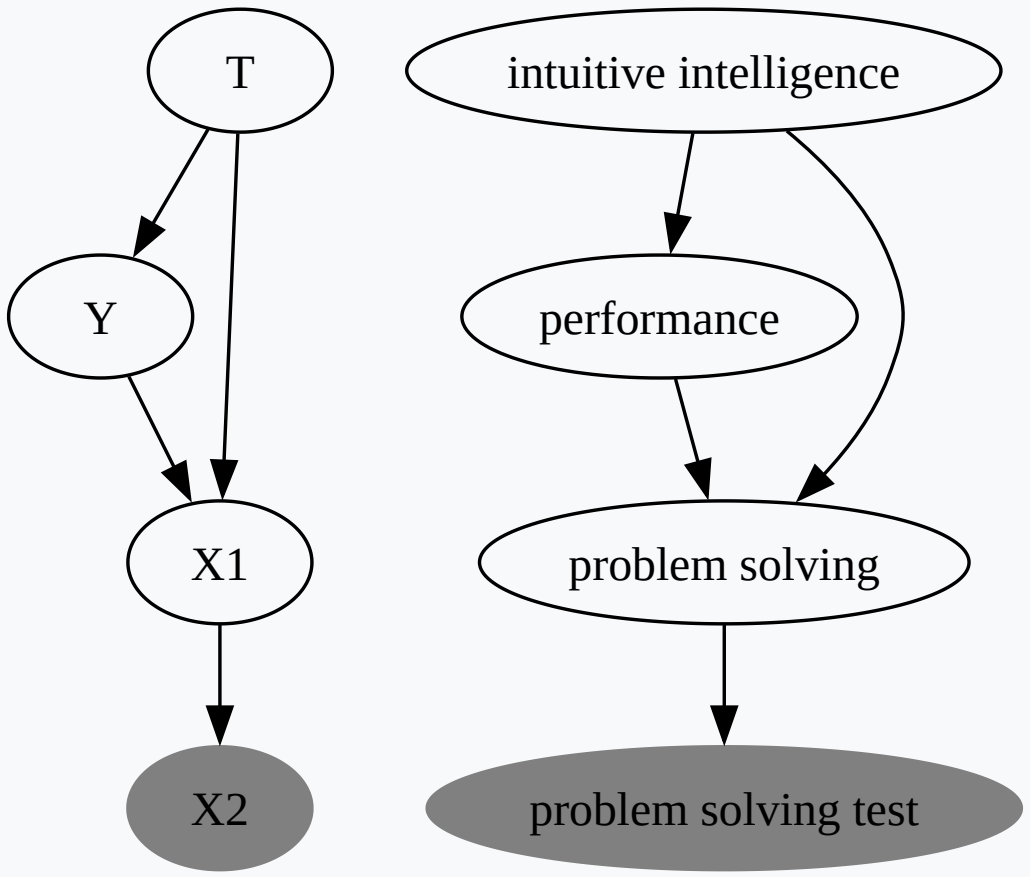
Selection Bias: Why Conditioning Hurts

`problem solving` is a **common effect** of `performance` and `intuitive intelligence`. By conditioning on it we introduce bias.

If we look at people with good problem-solving abilities, we are probably looking at people with either good performance or good intuitive intelligence. Knowing that a person has good performance **explains away** whether they have intuitive intelligence — so we see a correlation which does not represent a causal effect.

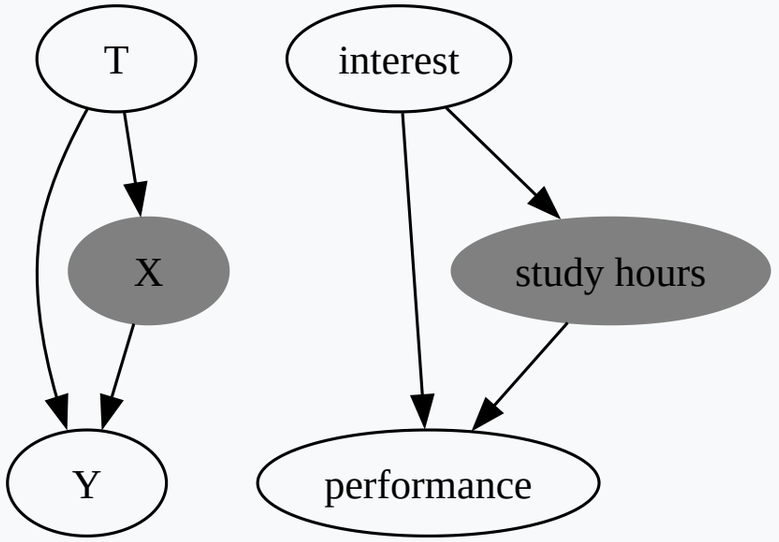
By conditioning on the common effect, we opened a backdoor path between Y and T which affects the perceived association between treatment and outcome.

The same is true if we condition on a **descendant** of the common effect:



Selection Bias: Controlling on a Mediator

Another case of selection bias is controlling for a **mediator** of treatment and outcome. We want the effect of `interest` on `performance`, and we could randomise interest (say, selecting students with different levels of interest at random across schools). **"Just to be sure", we condition on `study hours`:**

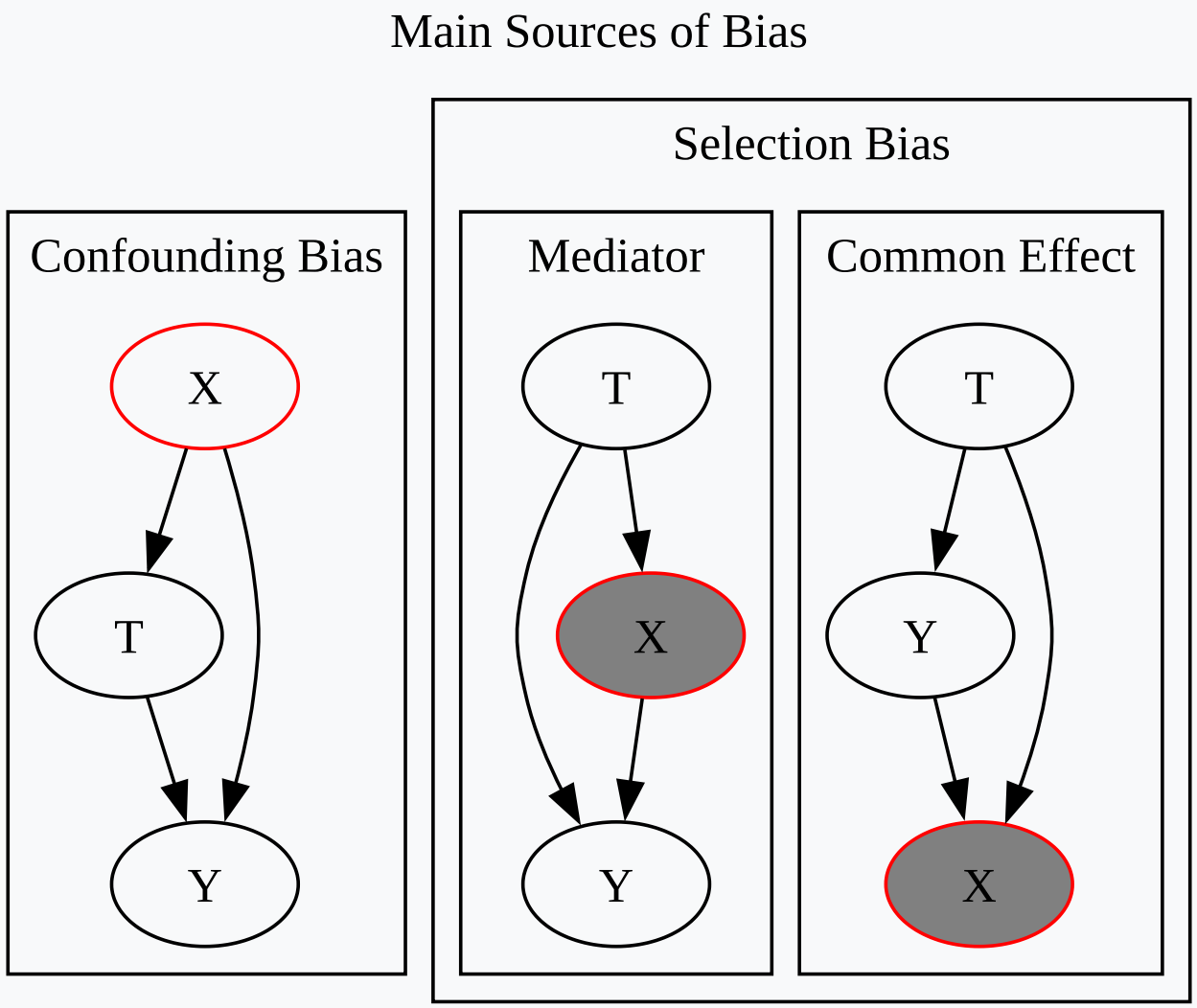


`study hours` lies on the path from the treatment to the outcome. Conditioning on the mediator **closes that path**, hence reducing the measured effect between `interest` and `performance`.

While we expect interested students to study more and perform better, **among people who study a lot, `interest` and `performance` may seem little correlated**: even people with little interest obtain good performance if they study for long hours.

Main Sources of Bias

To summarise, the main sources of bias are three:



Confounders

We **should** control on the common cause

Selection bias — common effect

We should **not** control on the common effect

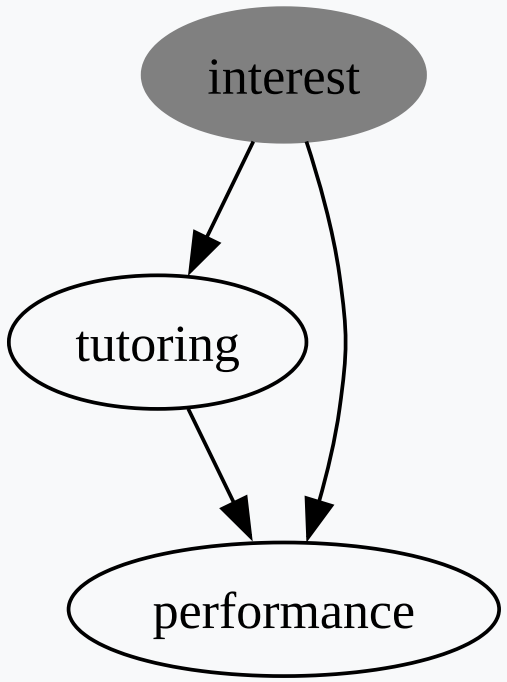
Selection bias — mediator

We should **not** control on the mediator

Red marks the nodes that are not conditioned on but should be, and the nodes that are conditioned on but should not be.

Linear Regression for Causal Analysis

When we said "we control for X" we meant: divide the data into groups, measure the association within each group, then aggregate the results into an average effect. **Linear regression automates it.** Take the tutoring example again, with `interest` as a possible confounder:



(the graph above, without conditioning, is the one on the "Graphical Causal Models" slide)

Recall the assignment is not random, so `interest` influences both `tutoring` and `performance`. If we ignore this bias we can compute the difference between treated and untreated:

$$E[performance|tutoring = 1] - E[performance|tutoring = 0]$$

and the average performance of the untreated, $E[performance|tutoring = 0] = 72.93$.

Both values are **automatically computed within a linear regressor** of the kind:

$$performance = \beta_0 + \beta_1 tutoring$$

Ignoring the Confounder

$$performance = \beta_0 + \beta_1 tutoring$$

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	72.9291	1.199	60.832	0.000	70.565	75.293
tutoring	**6.8269**	1.671	4.087	0.000	3.533	10.121

Coherently with the interpretation of linear regression:

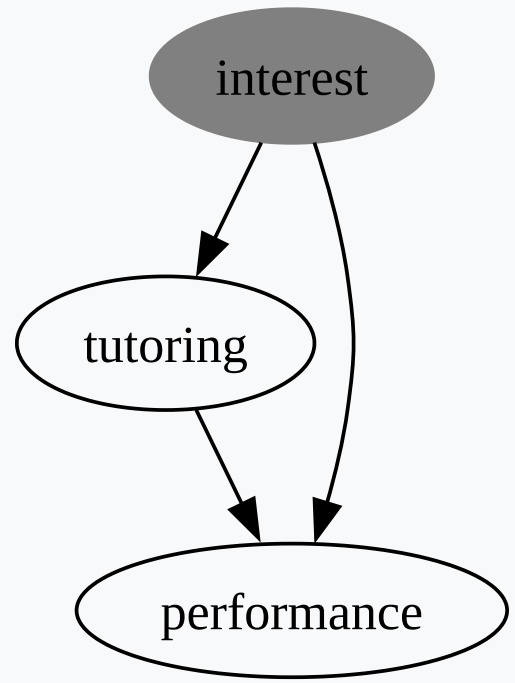
- `intercept`: the average performance of the **untreated** — the value we expect when `tutoring=0`;
- `tutoring`: the increment in `performance` when passing from `tutoring=0` to `tutoring=1` — the **average difference** between treated and untreated.

We also get confidence intervals and statistical tests **for free**.

There is a non-zero association: students who enroll in the tutoring programme increase their average performance by **6.83 points** (on a 0–100 scale). But we know this result is **biased** by the confounder `interest`. To remove the bias we should condition on it.

Controlling for `interest` in the Regression

The graph, with `interest` in gray:



One way to **control for a variable is to include it among the independent variables** of the regression:

$$performance = \beta_0 + \beta_1 tutoring + \beta_2 interest$$

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	54.1697	1.296	41.804	0.000	51.614	56.725
tutoring	**−0.9516**	1.129	−0.843	0.400	−3.178	1.275
interest	7.8097	0.441	17.711	0.000	6.940	8.679

The coefficients changed substantially. We are interested in the coefficient of `tutoring`:

> β_1 is the increment we expect to observe in `performance` when `tutoring` passes from 0 to 1 **and `interest` is held constant**.

What the Adjusted Coefficient Says

By including `interest` we are **effectively controlling for it**: the coefficient of `tutoring` now looks at a subset of the data in which `interest` is constant — the average effect of `tutoring` when comparing groups with constant `interest`.

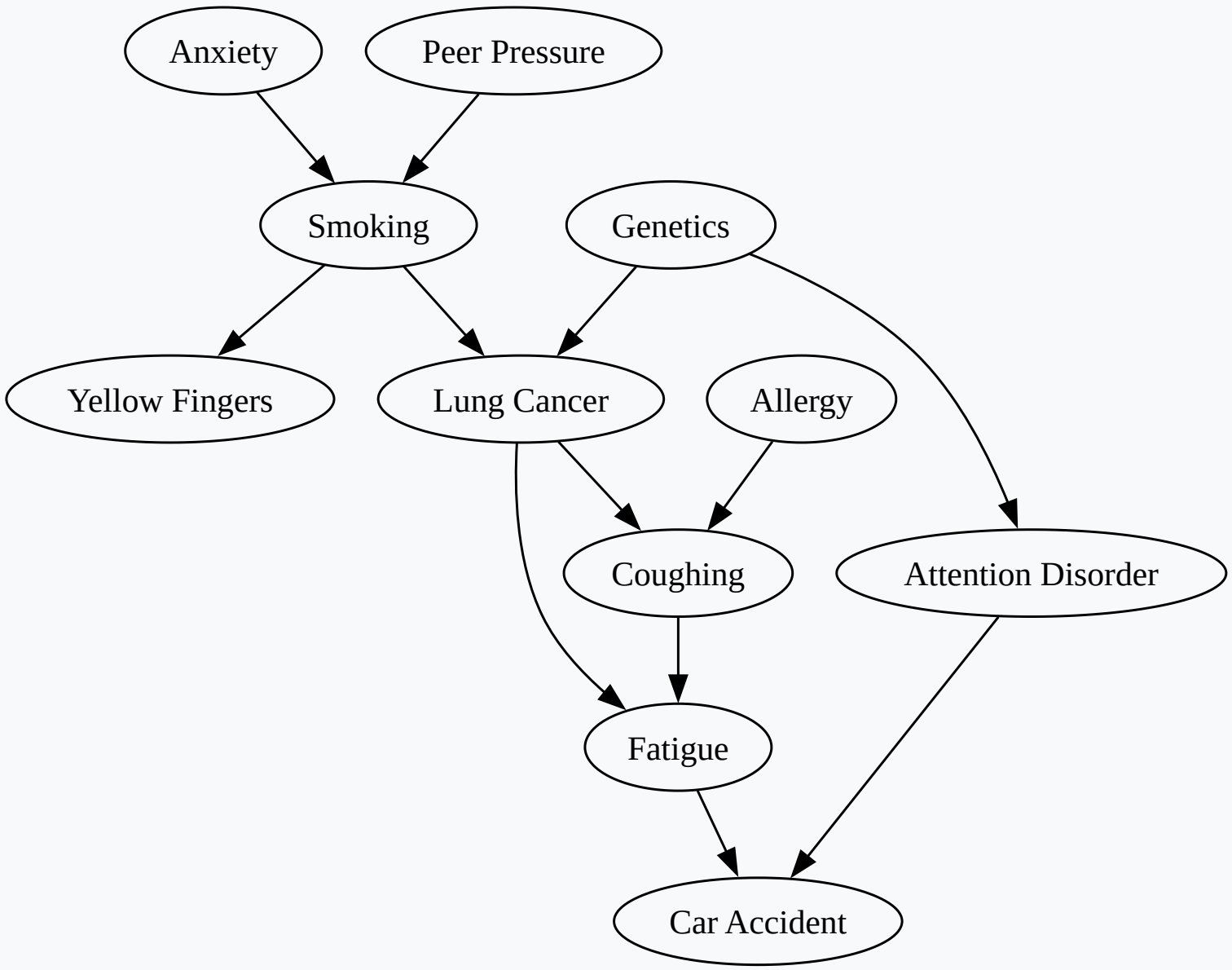
Hence, **even if we cannot physically perform a random assignment, linear regression lets us control for variables and analyse observational data**. This is fundamental: random tests are not always feasible.

The coefficient of `tutoring` has a p-value of 0.4. A value different from zero was computed, but the test tells us that **we do not have enough elements to say it is actually different from zero**.

- We can conclude that `tutoring` does **not** have a statistically significant impact on `performance` — the same conclusion reached with the randomized experiment, and with the groups split by `interest`.

Example on the LUCAS Dataset

LUCAS is an **artificial dataset**: values were drawn from known distributions to emulate a causal phenomenon. The authors propose the **true causal graph**, which reflects the way the data was generated — so we can check our reasoning against the truth.



The data has 2000 rows and binary variables: `Smoking`, `Yellow_Fingers`, `Anxiety`, `Peer_Pressure`, `Genetics`, `Attention_Disorder`, `Born_an_Even_Day`, `Car_Accident`, `Fatigue`, `Allergy`, `Coughing`, `Lung_cancer`.

Source: **causality.inf.ethz.ch/data/LUCAS.html**

Does `Smoking` Cause `Car Accident`?

Start by fitting a logistic regressor — all variables are binary, so the dependent variable needs it:

$$\log \frac{P(Car\ Accident|Smoking)}{P(\neg Car\ Accident|Smoking)} = \beta_0 + \beta_1 Smoking$$

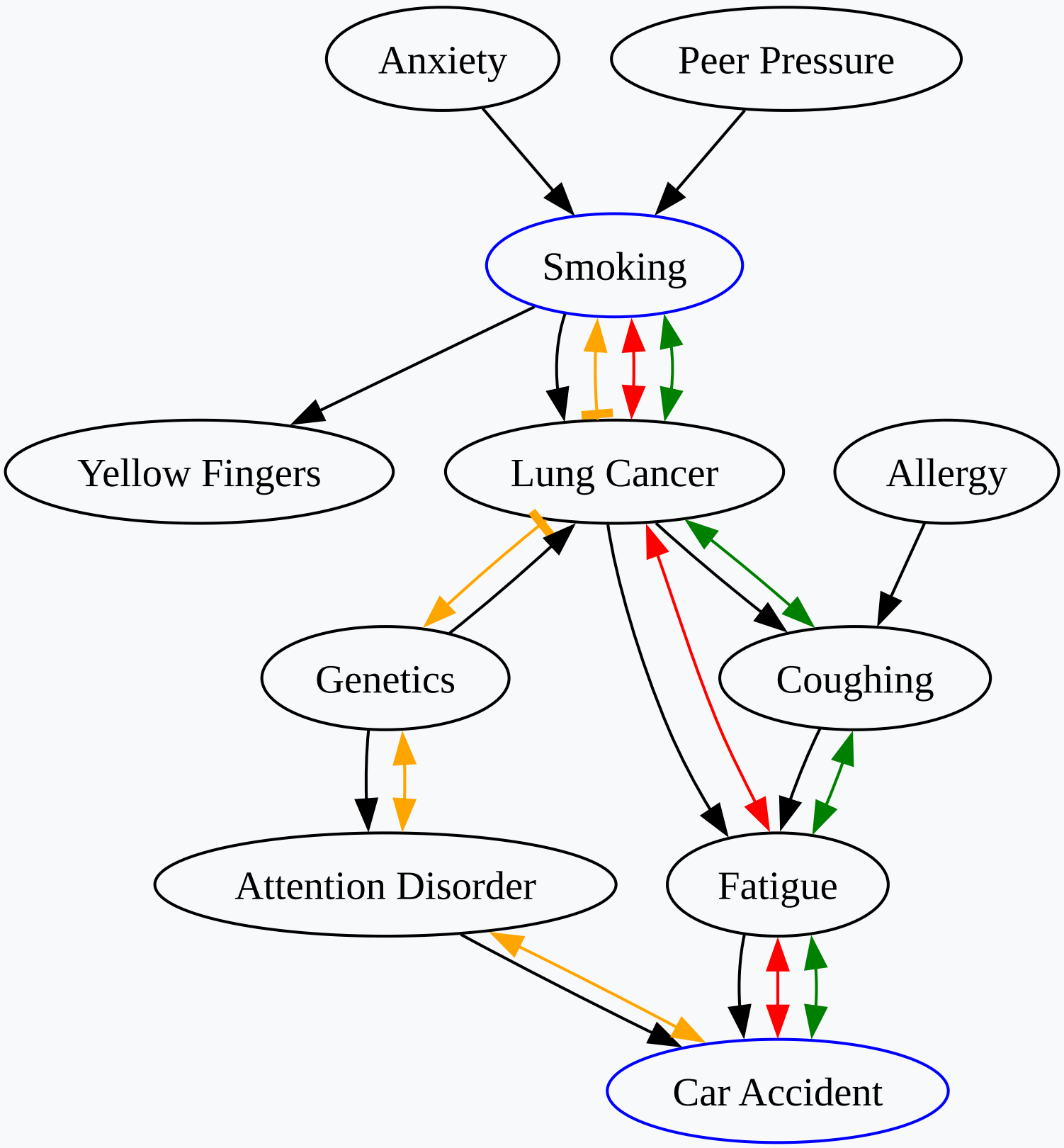
	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	0.7668	0.097	7.939	0.000	0.578	0.956
Smoking	**0.2596**	0.113	2.299	0.022	0.038	0.481

Since $e^{0.2596} \approx 1.3$, smoking is associated with a **+30%** increase in the odds of a car accident.

> But is this a causal effect? Does **smoking cause an increment in the odds of a car accident**?

We know there might be some bias. To check, we look at the graph and highlight **all dependence paths** from `Smoking` to `Car Accident`: these are all the different ways the two variables can be correlated.

The Paths Between `Smoking` and `Car Accident`

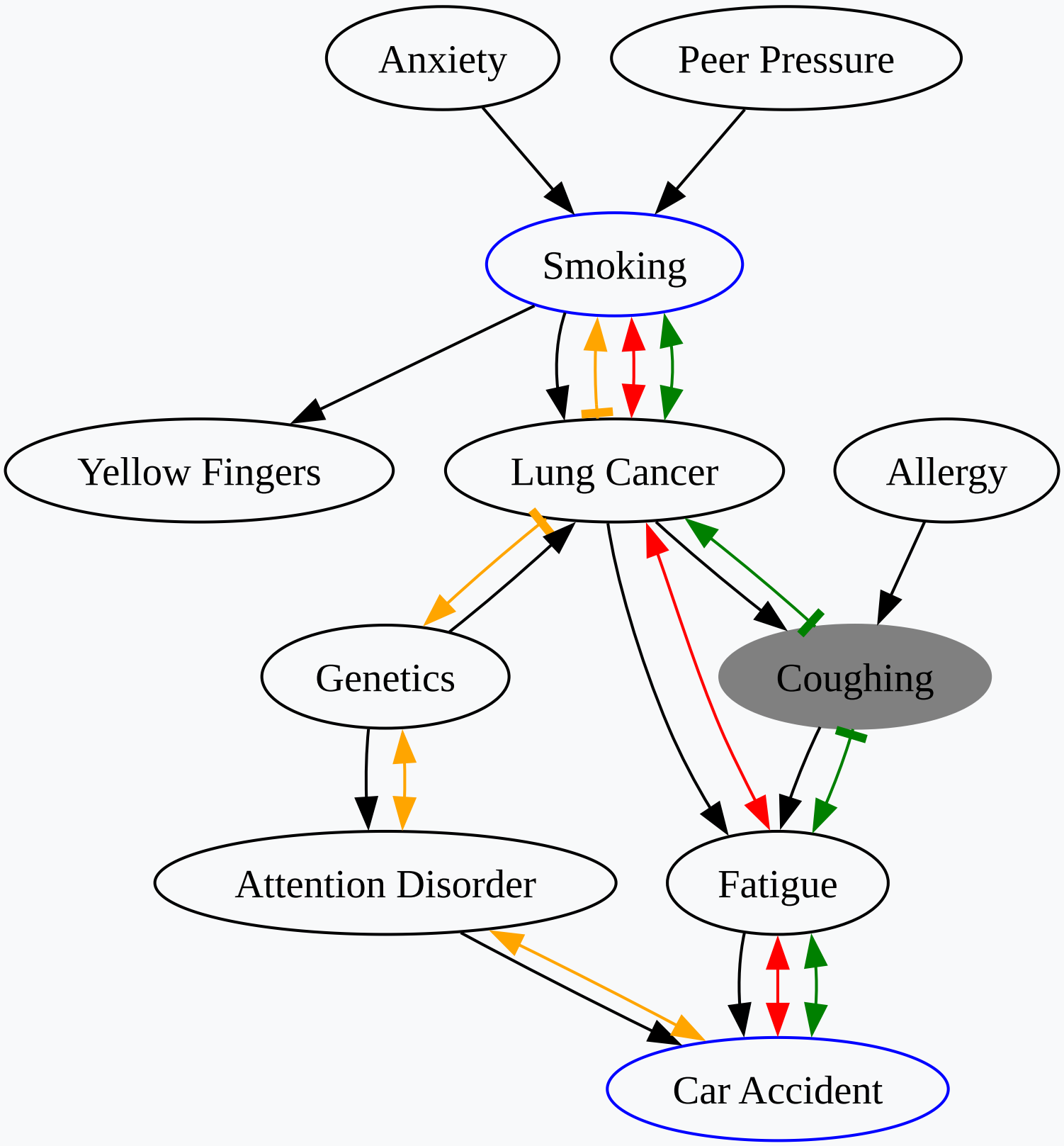


Three possible paths, in orange, red and green. Arrows indicate whether dependency flows or not:

- The **orange** path is **blocked**: there is a collider (`Lung Cancer`) that has not been conditioned on. The collider prevents `Smoking` and `Genetics` from being associated. This blocked path is fine: we are asking whether smoking causes car accidents, and since `Lung Cancer` does not cause `Genetics`, that path would not be a viable one to check for a causal effect.
- The **red** and **green** paths are **not blocked**. There are two ways `Smoking` can influence `Car Accident`: directly through `Fatigue`, or via `Coughing`. This may lead to a combined effect stronger than it should be.

Conditioning on `Coughing`

To estimate a causal effect we need to block one of the two open paths. The most sensible choice is to **condition on `Coughing`**, so that smoking can affect `Car Accident` only through `Lung Cancer` and then `Fatigue`:



To condition on `Coughing`, fit the logistic regressor:

$$\log \frac{P(Car\ Accident|Smoking, Coughing)}{P(\neg Car\ Accident|Smoking, Coughing)} = \beta_0 + \beta_1 Smoking + \beta_2 Coughing$$

`Smoking` and `Car Accident`: the Verdict

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	0.3411	0.108	3.153	0.002	0.129	0.553
Smoking	**−0.0130**	0.120	−0.108	0.914	−0.248	0.222
Coughing	0.9529	0.109	8.717	0.000	0.739	1.167

We now get a different picture: the p-value of `Smoking` is very large. **Once the influence of `Coughing` is removed**, the coefficient of `Smoking` is not significantly different from zero.

- This suggests that there is **no significant causal effect** between the two variables: the +30% we saw before was an association flowing through an open path, not an effect of smoking on accidents.

Conclusions and Next Steps



We Have Explored:

Potential outcomes and the treatment effect

Where the bias in a correlation comes from

Randomised controlled trials

Graphical causal models

Chains, forks and colliders

Confounding and selection bias

Regression as an adjustment device

- This closes the first block. In the next lectures we change goal: from explaining a relationship to predicting an outcome.

References

- Facure Alves, Matheus. *Causal Inference for the Brave and True*. **matheusfacure.github.io/python-causality-handbook**
- Pearl, Judea, and Dana Mackenzie. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.
- The LUCAS dataset: **causality.inf.ethz.ch/data/LUCAS.html**
- Lecture notes: **Chapter 11 — Causal analysis**